

System Design

Reflection 1

Rokas Pazera 551495

Design and Implementation

During weeks 1 and 2 I worked through the C4 model layer by layer. I started with the basic layers (context, container, component, code) step by step to make sure I truly understand how they are supposed to be designed. I had a short exercise where I had to classify example statements about a bookstore platform into the correct C4 layer. This exercise helped me understand differences between the layers in practical world.

Later in week 1, I made deployment and component diagrams. The component diagram shows how the software is split into parts inside the system. The deployment diagram shows which servers or devices those parts run on. Drawing both diagrams helped me see that the design is not finished until I decide where the software will actually run.

Week 2 was a bit harder than the first week. I built the full C4 set for a Task Manager application and set up a consistent color legend for every component. The hardest part was to find the right UML components UML in draw.io and making arrow connections between components. Other than that, it wasn't that challenging to draw the nodes or connections between them.

Collaboration

I did everything solo during these weeks. I relied on help from the slides and google.

Research and Professional Development

The biggest skill gap in these weeks was UML notation. C4 notation was pretty easy ("just use boxes and arrows"), so moving to UML stricter components and notations in assignment 4, it made me realize several things I didn't know precisely or forgot about them:

- The difference between associations (plain open arrow), dependency (dashed open arrow), and inheritance (solid line, triangle arrow) was not something I could confidently draw from memory before this period.
- I initially used mismatched arrow styles (mixing solid and dashed lines without a clear reason) across the same diagram. I only realized this mistake when I checked online for UML diagram examples and I was wondering, why are there different arrow/line types.

Progress and Insights

This is my first partial reflection, so there isn't any previous reflection to compare it with. The main insight from this period is that C4 and UML solve different problems: C4 is for quickly

showing architecture at different levels, while UML is precise enough to describe exact relationships between components.

A second insight is that a diagram is not finished with just components. It is finished when every component and arrow is connected with each other according to the requirements. Also, I must keep in mind the placement of each component, so that the arrows don't overlap when I start connecting them.

In the future, after adding components, I will not stop doing the diagram without adding all the relation arrows, coloring the components and making sure that everything is truly correct according to the assignment/code, so that I can start coding based on the diagram and I don't have to change anything in it.